

Engineering Reboot 2026

Week-02 Field Manual v3.0

Theme: Python Control Flow

Design Goal: Single-document execution package. Open this manual at 7:00 PM and complete the session without needing to consult other documents.

Day 06 – Conditionals

Mission Brief

Learn and apply conditionals through executable exercises and documented evidence collection.

Program File

age_classifier.py

Learning Objectives

- Understand the concept
- Implement the concept in Python
- Execute test cases
- Capture evidence
- Commit work to Git

Exercise: Age Classifier

Prompt exactly: Enter age:

Classification Rules:

0-12 = Child

13-19 = Teen

20-64 = Adult

65+ = Senior

Required Test Cases:

5 -> Classification: Child

13 -> Classification: Teen

35 -> Classification: Adult

70 -> Classification: Senior

Evidence Requirements

1. Source code screenshot
2. Test execution screenshots
3. Notes completed
4. Journal completed
5. Git status screenshot before commit (recommended)

Git Commands

git status

git add .

git commit -m "Day 06 Complete"

git push

Notes Template

Concepts Learned

Observations

Questions

Key Takeaways

Success Criteria

All exercises completed.

Expected outputs verified.

Evidence captured.

Git commit completed.

Git push completed.

Day 07 – Boolean Logic

Mission Brief

Learn and apply boolean logic through executable exercises and documented evidence collection.

Program File

eligibility_checker.py

Learning Objectives

- Understand the concept
- Implement the concept in Python
- Execute test cases
- Capture evidence
- Commit work to Git

Exercise: Eligibility Checker.

Prompt for age and citizenship status.

Output Eligible or Not Eligible based on defined rules.

Document minimum 4 test cases.

Evidence Requirements

1. Source code screenshot
2. Test execution screenshots
3. Notes completed
4. Journal completed
5. Git status screenshot before commit (recommended)

Git Commands

git status

git add .

git commit -m "Day 07 Complete"

git push

Notes Template

Concepts Learned

Observations

Questions

Key Takeaways

Success Criteria

All exercises completed.

Expected outputs verified.

Evidence captured.

Git commit completed.

Git push completed.

Day 08 – Loops

Mission Brief

Learn and apply loops through executable exercises and documented evidence collection.

Program File

counting_program.py

Learning Objectives

- Understand the concept
- Implement the concept in Python
- Execute test cases
- Capture evidence
- Commit work to Git

Exercise: Counting Program.

Display numbers 1 through 20 using a loop.

Display even numbers separately.

Capture screenshots of both outputs.

Evidence Requirements

1. Source code screenshot
2. Test execution screenshots
3. Notes completed
4. Journal completed
5. Git status screenshot before commit (recommended)

Git Commands

git status

git add .

git commit -m "Day 08 Complete"

git push

Notes Template

Concepts Learned

Observations

Questions

Key Takeaways

Success Criteria

All exercises completed.

Expected outputs verified.

Evidence captured.

Git commit completed.

Git push completed.

Day 09 – Loop Practice

Mission Brief

Learn and apply loop practice through executable exercises and documented evidence collection.

Program File

pattern_generator.py

Learning Objectives

- Understand the concept
- Implement the concept in Python
- Execute test cases
- Capture evidence
- Commit work to Git

Exercise: Pattern Generator.

Generate star patterns using nested loops.

Minimum patterns: square, right triangle, inverted triangle.

Evidence Requirements

1. Source code screenshot
2. Test execution screenshots
3. Notes completed
4. Journal completed
5. Git status screenshot before commit (recommended)

Git Commands

```
git status
git add .
git commit -m "Day 09 Complete"
git push
```

Notes Template

Concepts Learned

Observations

Questions

Key Takeaways

Success Criteria

All exercises completed.

Expected outputs verified.

Evidence captured.

Git commit completed.

Git push completed.

Day 10 – Mini Project

Mission Brief

Learn and apply mini project through executable exercises and documented evidence collection.

Program File

menu_driven_calculator.py

Learning Objectives

- Understand the concept
- Implement the concept in Python
- Execute test cases
- Capture evidence
- Commit work to Git

Mini Project: Menu Driven Calculator.

Operations: Add, Subtract, Multiply, Divide.

Menu repeats until Exit selected.

Document at least 10 test cases.

Evidence Requirements

1. Source code screenshot
2. Test execution screenshots
3. Notes completed
4. Journal completed
5. Git status screenshot before commit (recommended)

Git Commands

git status

git add .

git commit -m "Day 10 Complete"

git push

Notes Template

Concepts Learned

Observations

Questions

Key Takeaways

Success Criteria

All exercises completed.

Expected outputs verified.

Evidence captured.

Git commit completed.

Git push completed.